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RESEARCH ARTICLE

Automated Motor Tic Detection: A Machine Learning Approach

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ABSTRACT: Background: Video-based tic detection and scoring is useful to independently and objectively assess tic frequency and severity in patients with Tourette syndrome. In trained raters, interrater reliability is good. However, video ratings are time-consuming and cumbersome, particularly in large-scale studies. Therefore, we developed two machine learning (ML) algorithms for automatic tic detection.

Objective: The aim of this study was to evaluate the performances of state-of-the-art ML approaches for automatic video-based tic detection in patients with Tourette syndrome.

Methods: We used 64 videos of $n = 35$ patients with Tourette syndrome. The data of six subjects (15 videos with ratings) were used as a validation set for hyperparameter optimization. For the binary classification task to distinguish between tic and no-tic segments, we established two different supervised learning approaches. First, we manually extracted features based on landmarks, which served as input for a Random Forest classifier (Random Forest). Second, a fully automated deep learning approach was used, where regions of interest in video snippets were input to a convolutional neural network (deep neural network).

Results: Tic detection F1 scores (and accuracy) were 82.0% (88.4%) in the Random Forest and 79.5% (88.5%) in the deep neural network approach.

Conclusions: ML algorithms for automatic tic detection based on video recordings are feasible and reliable and could thus become a valuable assessment tool, for example, for objective tic measurements in clinical trials. ML algorithms might also be useful for the differential diagnosis of tics. © 2023 The Authors. *Movement Disorders* published by Wiley Periodicals LLC on behalf of International Parkinson and Movement Disorder Society.

Key Words: Tourette syndrome; machine learning; tic detection; deep neural networks; Random Forest; Face Mesh

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Gilles de la Tourette syndrome (GTS) is a common neurodevelopmental disorder defined by the presence of motor and vocal tics for more than a year with onset before the age of 18 years.[1] Prevalence is estimated to be 0.3% to 0.9%.[2] GTS is still a clinical diagnosis. There is no specific confirmatory laboratory, imaging, or genetic test. Valid tic detection, including the classification of movements as tics or not, requires clinical acumen.[3] Tic frequency and severity fluctuate considerably across shorter and longer periods,[4] rendering valid and sensitive clinical assessment of tics difficult. The most widely used tool to capture tic severity is the Yale Global Tic Severity Scale (YGTSS), a composite measure of subjective patients' and objective clinicians' assessments.[5] It is sensitive to change in GTS severity[6] and has been proven useful in clinical practice.[7] However, because it scores symptom severity within the last 7 days preceding the assessment, it cannot capture short-term changes, that is, changes occurring within less than 7 days. Furthermore, associations between YGTSS scores and objective video tic measurements have been reported to be low by some authors.[8] To this end, and as an objective measure of tic frequency and severity, the Modified Rush video rating scale (MRVRS) has become a useful tool.[9] Although it allows a very detailed, objective, independent, and blinded analysis of the patient's current tic state, the rating process is time-consuming and cumbersome. Because of these practical constraints, in spite of its usefulness, the MRVRS has not been used widely in clinical trials.[10,11] To facilitate objective tic assessment, an automated tic detection approach using machine learning (ML) based on video recordings is desirable. This would allow structured, time-efficient data acquisition and analysis. Of late, ML has been used broadly in the field of movement disorders.[12-14] However, previous work on automated tic detection is scarce, and data so far are limited and nonrepresentative. For instance, Wu and colleagues[15] developed a tic detection algorithm based on visual features and could show that pretraining a neural network on unlabeled image data could improve the results if only limited data of labeled videos are available. Barua and colleagues[16] used an accelerometer-based approach but included only data from healthy participants simulating tics. Bernabei and colleagues[17] also used sensor-based data derived from accelerometers attached to the trunk of patients with GTS and compared these data with the rating of simultaneously acquired video recordings. Accuracy, sensitivity, and specificity for tic detection were around 80%.[17] However, given a rostrocaudal distribution of tics,[18] recordings of truncal tics are not representative for tics in GTS. In the study of Cernera and colleagues, electromyography (EMG) and accelerometer data of voluntary movements were measured during video recordings following the Rush protocol. A support vector machine was then trained for each individual participant to classify tic and no-tic phases in the same individual, which yielded an accuracy across the different classifiers of 96.69% ± 4.84%.[19] This approach allows automated tic detection only on an individualized basis and requires EMG and accelerometers.

Against this background, the present study aimed to investigate the feasibility and validity to detect and precisely localize facial tics in videos of patients with GTS using ML, to identify factors that significantly influence the classification accuracy, and to examine the performance of the different models. As to the latter, we used two different nonindividualized, supervised ML models based on 1-second video snippets from 5-min video recordings. The first, more classical ML approach (referred to as Random Forest approach) involved feature engineering based on automatically detected and localized facial landmarks followed by a Random Forest classifier.[20] The other (referred to as deep neural network approach) relied on a neural network. Neural networks represent a fully automated approach, where features are learned directly from the data, rendering them particularly well suited for pattern recognition in large datasets.

Patients and Methods

Patients

Patients with GTS were recruited from our GTS specialty clinics in the Departments of Neurology and Paediatrics of the University Hospital Schleswig-Holstein, Campus Lübeck. We included 64 videos of 35 individuals (42 showing children [2 girls] and 22 showing adults [10 women] with GTS). The mean age was 12.6 ± 2.4 years in the children's group and 30.9 ± 10.6 years in the adult group. The diagnosis of GTS was based on the DSM-V (*Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition*) criteria for GTS.[1] Further details are provided in the Supporting Information Methods section in Data S1.

Data Acquisition

As a data source for tic detection, we used the video recordings taken during urge monitor recordings[21,22] because scoring of tics in urge monitor recordings entails a second-by-second tic detection and assessment required for the development of an automated tic detection, which the assessment of Rush video recordings does not include. Such second-by-second tic assessment was the basis for training the AI algorithms. The urge monitor, that is, urge-tic video acquisition, was performed according to a previous study[21] (for further details, see Supporting Information Data S1).

Two independent clinical raters rated each of the 5-minute-videos per second.[21] They were previously

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personally trained by one of the authors (A.M.). Given that non-tic movements are common in children, videos of children were prescreened by the two raters, and an agreement was made to determine which snippets most likely showed non-tic movements. Time intervals prescreened as non-tic behavior were included and rated with a tic intensity of 0 (see later). Next, the presence and intensity of motor tics in a range from 0 to 8 (see Supporting Information Table A.2 in Data S1) were noted per second in all videos (children and adults). Video seconds without a present motor tic were rated with the intensity rating 0. Vocal tics were not included, because we found in a previous study of our group that on the basis of Rush video recordings, motor tics rather than vocal tics are the best predictor to distinguish between patients with GTS and healthy control subjects.[23] In addition, the analysis of vocal tics would have required sound signal processing, which was not the focus of this study. In case of discrepant intensity ratings, the average of the tic intensity was used. All segments where the two raters disagreed regarding the presence or absence of a tic were reviewed again by A.M. He finally decided whether a tic was present in these segments and rated tic intensity. In total, 64 videos of n = 35 subjects were labeled. Each video had a duration of 5 minutes. We excluded one subject for not following the given instructions during data acquisition.

Of the remaining 34 subjects, 6 (12 videos of 3 children and 3 videos of 3 adults) were used to provide data for a validation set. These videos were not included in the training data and were used only to determine model hyperparameters, architectures, and optimal sets of features using Random Search. By means of a leave-one-subject-out cross-validation, the model was trained and evaluated using the data of the remaining 28 subjects (see overview of the distribution in Supporting Information Fig. A.1 in Data S1).

Automated Tic Detection

Two main state-of-the-art ML approaches for feature extraction were evaluated in this study, one based on handcrafted features computed on the landmark coordinates (Random Forest), and the other training a deep neural network to directly learn features from the video data (deep neural network). Note that in the Random Forest approach, a neural network was also used for landmark extraction, and the term refers only to the subsequent classification algorithm. An overview of the two approaches and their respective data preprocessing is shown in Figure 1. To allow application of the model to subjects not previously seen by the model without first adapting it to them, neither preprocessing of the data nor further analyses were dependent on the subjects.

Data and Preprocessing

In both approaches, we tracked facial landmarks. The identical input data with respect to participants and segments, as well as training and test splits, were used for both methods to ensure maximum comparability of results. Classification as tic or no-tic phase was based on 1-second video snippets with every movement associated with an intensity above a certain threshold being considered as part of the "tic" class. Due to the sampling rate of 30 Hz, one snippet contains $T = 30$ frames. We thus divided each 5-minute video into consecutive 1-second video snippets each comprising 30 images (frames). The windows are contiguous and do not overlap. This resulted in 300 video snippets in one 5-minute video. Clinical ratings per second were used as the corresponding labels. We selected a tic intensity threshold of 3 for our classification problem because interrater differences were too large for tic scores of 1 and 2 (see Table 1). Therefore, tics with an intensity rating of 3 or higher are

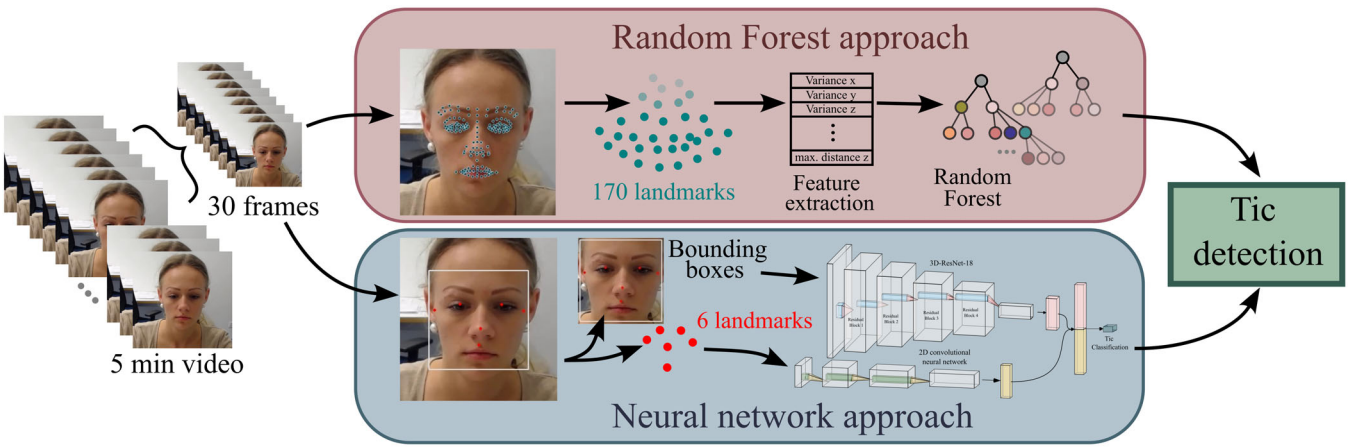

FIG. 1. Overview of the two proposed algorithms for tic detection. For both approaches, we extracted 30 frames from the videos as input. A total of 170 facial landmarks were extracted for Random Forest (red box), whereas face bounding boxes and six additional landmarks were used as input for the neural network (deep neural network, blue box). [Color figure can be viewed at wileyonlinelibrary.com]

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included. Using this threshold, the interrater agreement percentage is 97.4%, with a Cohen's kappa of 94.2. The proportion of 1-second video segments containing tics was 35.6%. On average, each video had 79.2 ± 56.5 (range, 0–215) segments containing tics.

Preprocessing for Random Forest

In the Random Forest approach, a set of landmarks was extracted from each frame of a video snippet, resulting in $T = 30$ landmark sets per snippet. They served as a lower-dimensional representation for the classifier. For landmark extraction, Google's MediaPipe Face Mesh[24] was used, which is based on a residual neural network (ResNet) returning 468 three-dimensional (3D) landmarks. In the next step, a subset of $L = 170$ landmarks important for the detection of tics was manually selected, and only these were used for further analysis. They included landmarks around the mouth, as well as contours of the eyes and nostrils (see Fig. 1; for further information, see Supporting Information Fig A.2 in Data S1). We did not include landmarks on body parts other than the face because they could not be extracted as reliably. We subtracted the coordinates of one central landmark (middle of the upper lip) from each landmark to center the data, and we applied augmentation by flipping the coordinates horizontally to make the model invariant to varying illumination. The final output of the described landmark extraction algorithm is an array C of dimension $3 \times L \times T$. The output regarding landmark k along axis d can be mathematically described as $C_d^{(k)} \in R^T$, where $d \in x, y, z$ and $k \in \{1, \ldots, 170\}$.

Preprocessing for Deep Neural Network

In the deep neural network approach, we created regions of interest as bounding boxes containing the subjects' faces for each frame of a video snippet. For this, we used the BlazeFace model from Bazarevsky et al.[25] An exemplary output of this face detection neural network model is shown in the red box in Figure 1. It contains the position, height, and width of the facial bounding box, as well as six landmarks on the nose tip, the ears, the eyes, and the lower lip. We found that the inclusion of these additional landmarks had a positive effect on training stability and classification accuracy and added only a few model parameters. Bounding boxes were calculated for each frame in a video snippet. Only the one with the highest confidence was used for cropping all 30 frames of one video snippet while all 30 landmark sets were used (see Supporting Information Fig. A.3 in Data S1). This procedure ensures that the facial landmarks and bounding boxes are predicted even in the case of missing outputs, for example, caused by occlusions, or in the case of temporarily misplaced bounding boxes.

The resulting images were resized to a resolution of 64×64 pixels. We applied different techniques to the image crops to augment the dataset, including random shifts of the crops, rotations up to an angle of seven degrees, shearing, horizontal flips, color distortions, and changes in saturation, brightness, and contrast. The x and y coordinates of the six additional landmarks were z score normalized by subtracting the mean and dividing by the standard deviation both computed on the coordinates of all frames contained in a snippet. As a result, the influence of the individual face shape is reduced but information about relative movements maintained.

Feature Extraction and Classification

Random Forest Approach

For the landmark-based approach, we manually extracted a set of features from each set of $3 \times L \times T$ landmark coordinates. We used the validation set to find the best subset of features for the classification. The features in the chosen subset consisted of the maximum Euclidean distance between the coordinates of two frames, the variance of the coordinates, as well as the mean and maximum of the absolute spatial drift of each landmark among two consecutive frames of one set of coordinates.

A Random Forest classifier[20] was applied to predict the presence or absence of a tic in a second using the previously computed features. Random Forests consist of several uncorrelated decision trees, which are grown under a certain kind of randomization during the learning process. For a classification, each tree within this forest makes a decision, and the class with the highest number of votes determines the final classification. We used Random Search on the validation set to find the best hyperparameters for the Random Forest classifier.[26] In addition, to investigate the importance of different landmarks to distinguish between tic and no-tic phases, we performed a SHapley Additive exPlanations (SHAP) analysis[27] (for further information, see Supporting Information Data S1).

Deep Neural Network Approach

A 3D ResNet-18[28] was used as the second model because of its suitability for video processing.

TABLE 1 *Total number of differing tic/no-tic ratings per score*

Rating	1	2	3	4	5	6	>6
Total no. of differing ratings	111	1230	261	27	15	6	0
Percentage of differing ratings	82.0	44.7	8.4	3.7	14.2	12.0	0.0

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It receives a red, green, blue video as input. For enhanced stability in training and improved classification results, we added a simple additional neural network branch, which receives six facial landmarks as input.

The batch size was set to 64 and like for the landmark-based Random Forest approach, we used video snippets of 1 second, resulting in $T = 30$ frames and landmark sets. Additional information on the neural network architecture and its training, as well as an illustration of the network architecture (Supporting Information Fig. A.4 in Data S1), is given in the Supporting Information.

To determine important facial regions for tic classification, gradient-weighted class activation mapping (GradCAM)[29] was used to generate heatmaps for the deep neural network approach. For methodological details, see Supporting Information Data S1.

Statistical Methods

We focused on the binary classification problem of tics against no tics. As classification metrics, we used the accuracy and the F1 score. The accuracy (ACC) is calculated by dividing the number of correct predictions by the total number of predictions and can be written as

$$\mathrm{ACC} = \frac{\mathrm{tp} + \mathrm{tn}}{\mathrm{tp} + \mathrm{tn} + \mathrm{fp} + \mathrm{fn}}.$$

The F1 score is defined by the harmonic mean of precision and recall. In our classification problem, precision is the number of correctly detected tics divided by the number of all detected tics. The recall, also known as sensitivity, is the number of detected tics divided by the overall number of tics. The F1 score, combining these metrics, is therefore also given by

$$\mathrm{F1} = \frac{\mathrm{tp}}{\mathrm{tp} + \frac{1}{2}(\mathrm{fp} + \mathrm{fn})}.$$

We calculated the metrics based on a leave-one-out validation. This involves using the data of each subject as a test dataset once and training a new model on the data of the remaining subjects.

To compare the results of both approaches, we calculated paired samples t tests.[30] We used Shapiro–Wilk tests[31] to test for normality. In case of a deviation from normality, we performed a Wilcoxon signed-rank test instead.[32] We also created the receiver operating characteristic (ROC) curves for the two classifiers (for details, see Supporting Information Data S1).

Results

Because the 1-second video segments were assigned to different rating scores from 0 to 8 and we defined a binary classification problem, it was necessary to choose a threshold value on the tic intensity above which an example is considered as belonging to the tic class. We calculated the accuracy and the F1 score for different thresholds (see Supporting Information Table A.3 in Data S1).

With higher threshold values, the accuracy increases, whereas the F1 score peaks at a tic intensity of 3 or greater. We chose a threshold of 3 to define tics for our classification, including tics with an intensity rating of 3 or greater. Using the data from all 28 cross-validation steps, we calculated the mean accuracy, the mean F1 score, and the 95% confidence bounds of accuracy and F1 score for both approaches. For this specified classification problem, the Random Forest approach based on landmarks achieved an F1 score of 82.01% (95% confidence interval [CI]: 77.39, 86.63) and an accuracy of 88.44% (95% CI: 85.11, 91.77), respectively, whereas the fully automated neural network approach (deep neural network) yielded an F1 score of 79.49% (95% CI: 75.27, 83.74) and an accuracy of 88.50% (95% CI: 86.33, 90.66). A subject-specific overview of the accuracies of both approaches can be found in Supporting Information Figure A.5 in Data S1.

The F1 score and accuracy distributions are displayed in Figure 2 as boxplots. Shapiro–Wilk test showed that the differences of F1 scores ($W = 0.968$, $P = 0.539$), as well as the differences of accuracies ($W = 0.939$, $P = 0.105$), of the two approaches followed a normal distribution. The paired samples t test showed that the F1 scores [$t(27) = 0.924$, $P = 0.364$] and the accuracies [$t(27) = -0.03$, $P = 0.976$] did not differ significantly between the proposed approaches.

Furthermore, we calculated the ROC curves for all subjects and both approaches. Figure 3 shows the overall mean ROC curve, as well as the mean ROC curves across all subjects. The areas under the curve were 0.94 for the Random Forest classifier and 0.92 for the deep neural network. This small difference was significant (paired samples t test: $t(27) = -2.442$, $P = 0.021$).

A SHAP analysis indicated that the feature with the highest impact on the model's decision was the maximum Euclidean distance between the coordinates of two frames. The mouth region had the greatest influence on the model's prediction. The Supporting Information includes detailed results of the SHAP analysis (Supporting Information Figs. A.6, A.7, and A.8 in Data S1).

Gradient-Weighted Class Activation Mapping

We inspected a variety of the attention maps created with GradCAM for video snippets correctly classified as tics. We observed that in the videos where, for example, eye or mouth tics occurred, the corresponding facial regions were also assigned high importance. The

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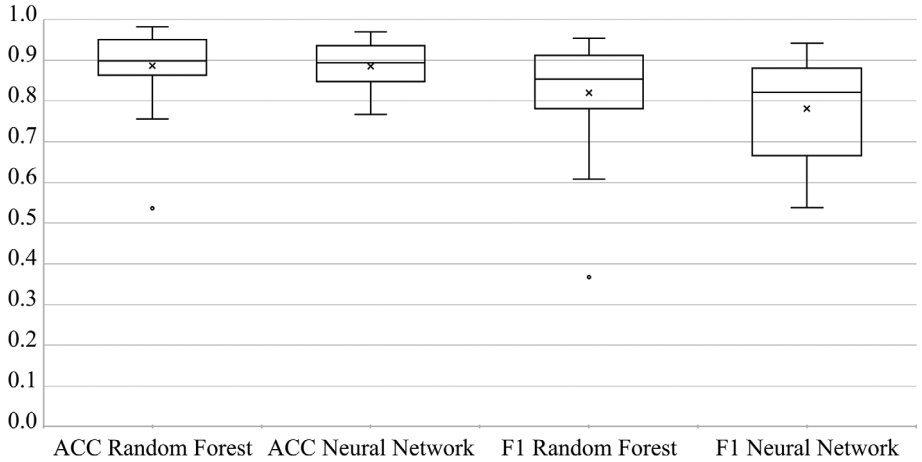

FIG. 2. Box plots for the visualization of the distribution of the results of accuracy (ACC) and F1 score (F1) of both applied machine learning approaches. Markers indicate mean values. Only tics with intensities >3.0 were considered as part of the tic class.

adequate resolution of the heatmap allowed discrimination of nearby tic regions such as those involving the eyes or eyebrows, and the temporal resolution enabled determination of the exact timing of the tic. Furthermore, it was apparent that videos misclassified as tics often contained blinking. Segments where tics were erroneously not detected often included mild mouth, nose, or eye tics, such as pulling up one corner of the mouth or nasal flare. An example of GradCAM is included in the Supporting Information (Fig. A.9 in Data S1).

Discussion

In this study, we applied two different ML approaches to the problem of binary classification between tics and no tics in video recordings. We show that the established ML approaches were successful in classifying video segments into tic or no-tic phases with high F1 score up to 82%, accuracy of up to 89%, and area under the curve of up to 94%, enabling automatic tic detection. The results of the two models were similar, although the Random Forest classifier appeared to be slightly superior. These findings can be taken as a basis for further developing our ML approaches into applications allowing to distinguish between people with tics and those without.

For the tic classification, we used different tic intensity thresholds. We found that the classification accuracy of both approaches increased with the use of higher thresholds. This is due to the definition of the tic score, where a tic score of 3 is defined as noticeable "to any person attending to it," and a tic with an intensity less than 3 "may be noted by others." By focussing on

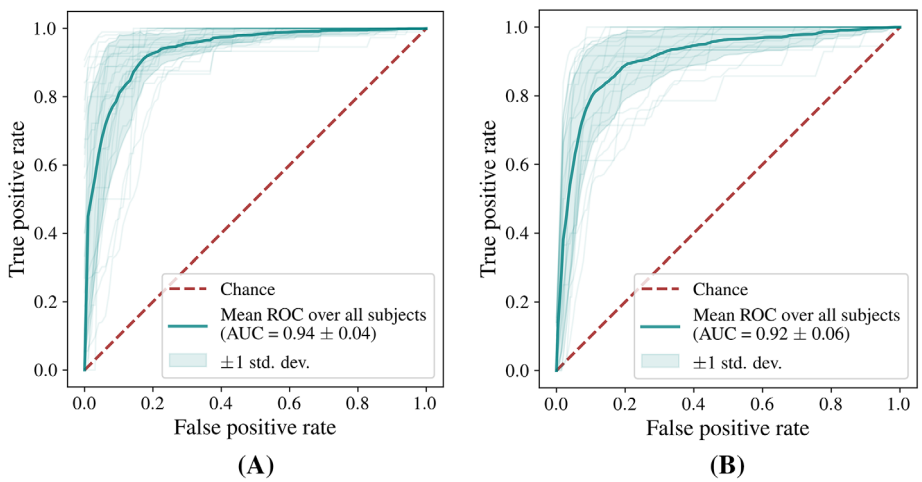

FIG. 3. Receiver operating characteristic (ROC) curves for the Random Forest (**A**) and deep neural network approach (**B**). The bold curve is the mean over the ROC curves of all subjects that are shown in light blue. The light blue background shows a range of 1 standard deviation (std. dev.). For reference, we have included the ROC curve of a random classifier (red line). AUC, area under the curve. [Color figure can be viewed at wileyonlinelibrary.com]

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the clearly distinguishable tics, the ML algorithms performed better (see Supporting Information Table A.3 in Data S1).

It is difficult to compare the accuracy of the few existing tic-detection approaches used in other studies with ours because methods differed considerably.[15-17,19,33] For instance, in one study, a neural network was trained on the basis of movements of healthy control subjects who simulated motor tics.[16] Cernera and colleagues[19] successfully developed an algorithm discerning tics and voluntary movements in tic patients using wearable sensors, with training being subject specific. Bernabei and colleagues[17] used accelerometers attached to the trunk in patients with GTS and controlled this using video recordings. In this study, we used video material, which can easily be recorded, accessed, and interpreted without additional sensors being necessary. In another study, Tang and colleagues[33] used videos from a single subject for facial tic detection. In contrast with previous studies, our models were trained across subjects and included data from children and adults, increasing their clinical usefulness and applicability (see Supporting Information Data S1 for subgroup comparison and generalization).

Comparing our two approaches, both have strengths and weaknesses. Although the Random Forest approach is not a common choice for more complex problems such as image or video classification, it performs well in cases with limited training data. Moreover, it allows to incorporate prior knowledge through the choice of features used for the classification, so that interpretation of the results is facilitated. We could demonstrate that features derived from facial landmarks of video frames are useful and reliable as input for a Random Forest classifier, with the mouth region having the greatest influence on the model's decision. Deep neural networks usually require a larger set of training data. In contrast with the Random Forest approach, no prior knowledge is needed because the neural network learns the features directly from the input. In this study, we show that the deep neural network approach is also applicable to the given problem with limited data. An advantage of this approach is that it can be used to highlight regions and points in time, which are important for the classification decision via GradCAM. The latter could be useful as a tool for training raters or for verifying the plausibility of decisions.

Both of our methods have been used for the detection of other movement disorders, for instance, Parkinson's disease, where ML has been widely used. For example, Açıcı and colleagues[34] investigated wearable sensors for gait analysis, and Polat[35] examined voice detection, both using a Random Forest classifier. Regarding deep neural networks, Aversano and colleagues[36] discriminated patients with Parkinson's disease and healthy control subjects using feet sensors for gait analyses; Shivangi et al[37] proposed an image classification approach.

A limitation of this study and other comparable studies is that the proposed approaches depend on the correctness and accuracy of clinical tic ratings. To minimize this inherent problem, we combined ratings of trained raters and had uncertain sections reviewed again by a third expert. In addition, it should be noted that although the intensity scale has been used successfully in previous studies,[21,22] it has not been formally validated. Also, based on the 1-second video snippets, it is not always possible even for experts to decide whether a tic is present because of the lack of long-term contextual information. It should also be noted that many of the Random Forest features and the output of the proposed neural network are dependent on the frame rate of the video recordings. Therefore, when applying the model, it is important to keep the frame rate at around 30 Hz to ensure that the differences in coordinates between successive frames are comparable with those in the training data.

The next step could be to use ML to distinguish tics in GTS from spontaneous movements in healthy control subjects based on video recordings of both groups. MRVRS recordings could be used for this purpose. On this basis, the algorithm could generate Rush scores and tic counts per minute. These in turn could be used as a clinical tool to automatically assess the intensity and frequency of tics, for example, to monitor treatment outcomes under real-life conditions or as part of clinical trials. In contrast with the YGTSS, short-term changes could be captured, for example, severity changes during the day. A further refined ML algorithm could also facilitate the distinction between tics and functional tic-like behavior, which has become a relevant problem in clinical practice.[38,39] It is worth noting that based on the proposed model, it is possible to also develop a mobile application for these purposes, because both the landmark extraction model (Random Forest classifier) and the bounding box model (deep neural network) can be implemented on mobile devices. Building on this, it is conceivable to use a Random Forest classifier or a real-time capable neural network comparable with the network used in this study to ensure classification and assessment with a minimum delay. This could be used to develop an at-home tic evaluation app. In turn, this would provide an advantage compared with the original Rush setting; that is, the speed, availability, and frequency of the evaluation would be a lot higher, and the inconvenience for patients to repeatedly travel to specialist clinics could be reduced.

To conclude, this study shows that motor tic detection is feasible and reliable, particularly for easily noticeable tics; that is, those with an intensity score >3. Both approaches tested appear useful, with the Random

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Forest classifier being particularly useful for limited datasets and allowing to incorporate prior knowledge about tics. The data provided in this article can be taken as a starting point to further develop ML approaches for tic detection into clinically useful tools, for example, as outcome measures in clinical trials and aids for differential diagnosis.

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Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Data

Additional Supporting Information may be found in the online version of this article at the publisher's web-site.

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